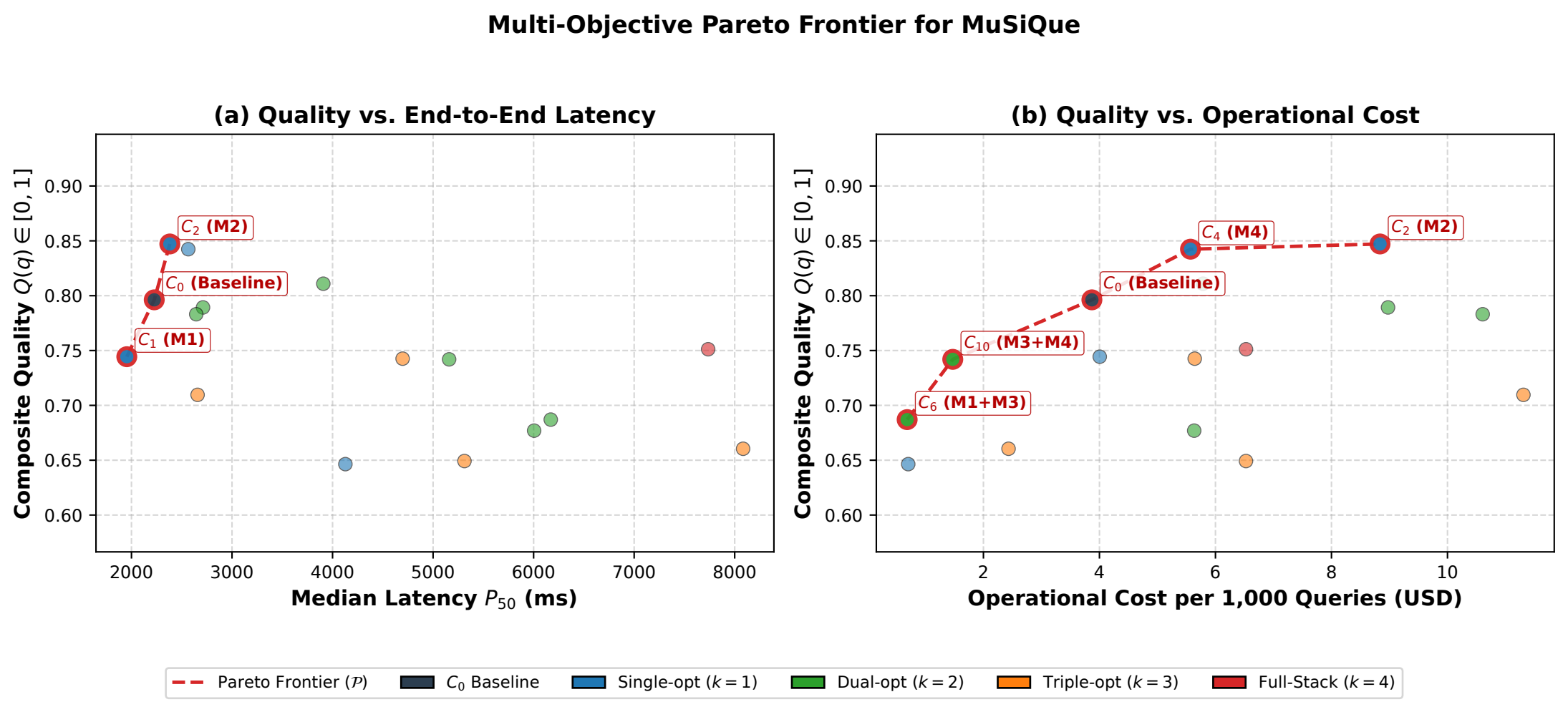
Multi-Objective Pareto Frontier for MuSiQue
(a) Quality vs. End-to-End Latency
(b) Quality vs. Operational Cost
Composite Quality $Q(q) \in [0, 1]$
Median Latency P_{50} (ms)
Operational Cost per 1,000 Queries (USD)
C_2 (M2)
C_0 (Baseline)
C_1 (M1)
C_4 (M4)
C_{10} (M3+M4)
C_6 (M1+M3)
Pareto Frontier ($\mathcal{P}$)
C_0 Baseline
Single-opt ($k = 1$)
Dual-opt ($k = 2$)
Triple-opt ($k = 3$)
Full-Stack ($k = 4$)